t3-w06 — condition C0

Construction — Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

> The curriculum objective alone. No lesson plan, no note.

INPUT LAYERS

Input layer	What it carries	Channel	In this condition
Scope	what is to be taught, and to whom	prompt	**present**
Prior knowledge	what the learner already knew coming in	prompt	absent
Materials	what the lesson was represented with	prompt	absent
Position	where this objective sat in the sequence	prompt	absent
Method	the shape of the teaching, as event labels	prompt	absent
Enacted content	what was actually explained and worked through	prompt	absent
Set work	what the learners were asked to do in class	prompt	absent
Assessment	the questions the lesson was checked with	prompt	absent
Figures	the drawings the lesson used	tool	absent

WHAT THE MODEL IS SENT

Contract `ComposeWriteObjectiveSection`, declaring: `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Construction

`subtopic`

Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

`term`

Third Term

`week`

6

`lesson_objectives`

• construct parallel lines
• construct perpendicular lines
• bisect a given line segment
• construct angle 90°
• construct angle 60°

Objective 1 — construct parallel lines

`objective_text`

construct parallel lines

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 2 — construct perpendicular lines

`objective_text`

construct perpendicular lines

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 3 — bisect a given line segment

`objective_text`

bisect a given line segment

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 4 — construct angle 90°

`objective_text`

construct angle 90°

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 5 — construct angle 60°

`objective_text`

> construct angle 60°

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

WHAT THE MODEL PRODUCED

Objective 1

How to Construct Parallel Lines

> EXPLANATION
> To draw a line that is parallel to a given line, you need a ruler and a set-square (or a drafting triangle).
>
> 1. Place the set-square so that one of its edges lies exactly on the given line.
> 2. Keep the set-square steady and slide the ruler along the other edge of the set-square.
> 3. Draw a short line segment with the ruler. This segment is parallel to the original line because it follows the same direction set by the set-square.
> 4. Extend the segment as far as you need – it will stay parallel to the original line.

> WORKED_EXAMPLE
> Example: Construct a line parallel to line AB through point C.
>
> - Place the set-square so that one side rests on AB.
> - Hold the set-square steady and slide a ruler along the opposite side.
> - From point C, draw a short line segment CD using the ruler.
> - Extend CD; it is the required line parallel to AB.

1. Construct a line parallel to the given line PQ through the point R. Describe the steps you would take.
answer: 1. Place the set-square with one edge on PQ. 2. Slide a ruler along the other edge of the set-square. 3. From point R, draw a short segment RS using the ruler. 4. Extend RS; it is parallel to PQ.

2. You have a line CD and need a line through point E that is parallel to CD. What tool do you use and how?
answer: Use a set-square. Place one edge on CD, slide a ruler along the opposite edge, draw a short segment from E, then extend it. The new line is parallel to CD.

3. Why will the line you draw using the set-square be parallel to the original line? Explain in one sentence.
answer: Because the set-square keeps the ruler moving in the same direction as the original line, maintaining equal angles throughout.

How to Construct a Perpendicular Line

EXPLANATION

A perpendicular line meets another line at a right angle (90°). To draw a perpendicular line through a given point, you need a compass and a straightedge. The steps are:

1. Place the compass point on the given point and draw an arc that cuts the original line at two points.
2. Without changing the compass width, place the point on each intersection and draw two arcs that intersect above (or below) the line.
3. Join the given point to the intersection of the two arcs with a straightedge. The new line is perpendicular to the original line.

WORKED_EXAMPLE

Example: Construct a line perpendicular to line AB through point C.

- Step 1: With the compass on C, draw an arc cutting AB at D and E.
- Step 2: Keep the same radius. With the compass on D, draw an arc above AB. With the compass on E, draw another arc that meets the first one at point F.
- Step 3: Draw a straight line through C and F. Line CF is perpendicular to AB.

1. Construct a line perpendicular to the given line PQ through point R. Describe the steps you would follow.

answer: 1. Place compass on R, draw an arc cutting PQ at S and T. 2. Keep the same radius. With compass on S draw an arc above PQ; with compass on T draw another arc meeting the first at U. 3. Draw line RU. RU is perpendicular to PQ.

2. You have a line XY and a point Z not on the line. After drawing the first arc, you notice it only cuts XY at one point. What should you do?

answer: Adjust the compass to a larger radius so the arc cuts XY at two points, then continue with the construction.

3. Why does the line you draw through the given point and the intersection of the two arcs form a right angle with the original line?

answer: Because the arcs create a circle centered at the given point, and the points where the arcs intersect are equidistant from the two intersection points on the original line, forming an isosceles triangle whose base is the original line segment. The line through the given point and the arc intersection is the perpendicular bisector of that segment, giving a 90° angle.

How to Bisect a Line Segment

> EXPLANATION
> To bisect (cut exactly in half) a line segment you need a compass and a straightedge.
>
> 1. **Draw the segment** – label the ends A and B.
> 2. **Place the compass point on A** and open it to a radius **greater than half the length of AB**.
> 3. **Swing an arc** above and below the line.
> 4. **Without changing the radius**, place the compass point on B and draw another arc that cuts the first two arcs.
> 5. The two intersection points of the arcs give you a line.
> 6. **Draw a straight line** through these intersection points.
> 7. The point where this line meets AB is the **mid-point M**, and AM = MB.
>
> This construction works because the arcs ensure that any point on the new line is equidistant from A and B, so the crossing point on AB must be exactly halfway between them.

> WORKED_EXAMPLE
> **Example**
>
> You have a segment CD that is 8 cm long.
>
> 1. Place the compass on C and open it to 5 cm (more than half of 8 cm).
> 2. Swing an arc above and below CD.
> 3. Without changing the compass width, place the point on D and swing another arc that cuts the first two arcs.
> 4. Mark the two intersection points P (above) and Q (below).
> 5. Draw the line PQ. It meets CD at point M.
> 6. Measure CM and MD – each will be 4 cm, confirming that M bisects CD.
>
> So, the midpoint M divides CD into two equal parts of 4 cm each.

1. A line segment EF is 10 cm long. Using the bisecting method, what is the length of each half after you find the midpoint?
answer: 5 cm

2. You bisect a segment GH and obtain the midpoint K. If GK measures 7 cm, what is the total length of GH?
answer: 14 cm

3. A segment MN is 12 cm long. After constructing the bisector, you measure the distance from M to the midpoint P and find it to be 6 cm. Verify that P indeed bisects MN.
answer: Yes, because both MP and PN are 6 cm, so the segment is divided into equal halves.

How to Construct a Right Angle (90°) with a Compass and Straightedge

EXPLANATION

A right angle measures exactly 90 degrees. You can draw one using only a ruler and a compass. The method works for any point where you need the right angle:

1. **Mark the vertex** – Choose the point O where the right angle will be.
2. **Draw a base line** – With the ruler, draw any straight line through O. Call the part to the right of O point A.
3. **Set the compass width** – Place the compass point on O and open it to any convenient radius.
4. **Draw an arc** – Swing an arc that cuts the base line at point B.
5. **Create a second arc** – Without changing the compass width, place the point on B and draw another arc that intersects the first arc. Call the intersection point C.
6. **Draw the perpendicular line** – Use the ruler to join O and C. The line OC is perpendicular to the base line OA, forming a 90° angle at O.

Why it works: The two arcs create a circle with centre O and another with centre B. The line OC is the line joining the centres of two circles that intersect at C, which is the radius of the first circle. This geometry guarantees that ∠AOC is a right angle.

WORKED_EXAMPLE

Worked Example

Goal: Construct a right angle at point P on a piece of paper.

1. Draw a straight line through P and mark a point Q to the right of P.
2. Place the compass point on P, open it to about 3 cm, and draw an arc that cuts the line at R.
3. Without changing the width, move the compass to R and draw a second arc that meets the first arc at S.
4. Join P and S with a ruler.

The angle QPS is now a right angle. You can check by measuring with a protractor – it should read 90°.

Key points: The same compass width is used for both arcs, and the line joining the original point (P) to the intersection of the arcs (S) is the perpendicular line.

1. Construct a right angle at point X on a fresh sheet of paper. Describe the steps you would follow.
answer: 1. Draw any straight line through X and mark a point Y on it. 2. Set the compass at X, choose a convenient radius, and draw an arc cutting the line at Z. 3. Keep the same radius, place the compass at Z and draw another arc intersecting the first arc at W. 4. Draw the line XW. The angle YXW is a right angle.

2. You need a right angle at point M to start drawing a square. Using only a ruler and compass, how do you create the 90° angle?
answer: 1. Draw a line through M and mark a point N on it. 2. With the compass at M, draw an arc that meets the line at P. 3. Without changing the compass width, place it at P and draw a second arc intersecting the first at Q. 4. Connect M to Q with a ruler. The angle NMQ is 90°.

3. Explain why the line drawn from the original point to the intersection of the two arcs is always perpendicular to the base line.
answer: The two arcs are parts of circles with equal radii. The intersection point is equidistant from the original point and the point where the first arc meets the base line. This creates an isosceles triangle with the base as the original line segment, and the line joining the two centres is the perpendicular bisector of the base, giving a 90° angle at the vertex.

Objective 5

How to Construct a 60° Angle

> EXPLANATION
> A 60° angle can be made by drawing an equilateral triangle. An equilateral triangle has all three interior angles equal to 60°, so any two sides meeting at a vertex give you the required angle.
>
> **Steps**
> 1. Draw a straight line segment AB of any convenient length.
> 2. Place the compass point on A and draw an arc above the line.
> 3. Without changing the compass width, place the point on B and draw another arc that cuts the first one at point C.
> 4. Join AC and BC with straight lines. Triangle ABC is equilateral, and the angle ∠ABC (or ∠BAC, ∠ACB) is 60°.
>
> You now have a precise 60° angle at any of the triangle's vertices.

> WORKED_EXAMPLE
> **Worked Example**
>
> Suppose you need a 60° angle at point D on a line segment DE.
>
> 1. Draw DE.
> 2. With the compass set to the length of DE, place the point on D and draw an arc above the line.
> 3. Keeping the same radius, place the compass point on E and draw a second arc intersecting the first at F.
> 4. Connect D to F and E to F. The angle ∠DFE is 60°, and the angle at D (∠FDE) is also 60° because triangle DFE is equilateral.
>
> You can now use the ray DF as one side of a 60° angle starting from D.

1. Construct a 60° angle on a line segment XY that is 8 cm long. Describe the steps you would follow.
answer: 1. Draw XY (8 cm). 2. Set compass to 8 cm. 3. With point on X, draw an arc above XY. 4. With point on Y, draw another arc intersecting the first at Z. 5. Join XZ and YZ. Triangle XYZ is equilateral, so ∠XYZ (or ∠XZY, ∠ZXY) is 60°.

2. You have a line segment AB. Using only a ruler and compass, create a 60° angle with its vertex at A, without drawing the full triangle. What is the shortest sequence of actions?
answer: 1. With compass set to any convenient length, place point on A and draw an arc. 2. Without changing the radius, place compass on a point C on the arc and draw a second arc intersecting the first at D. 3. Draw line AD. The angle between AB and AD is 60° because triangle ACD is equilateral.

3. Why does constructing an equilateral triangle guarantee a 60° angle? Explain in one sentence.
answer: Because the interior angles of an equilateral triangle are all equal and sum to 180°, so each must be 180°/3 = 60°.

LAYER BY LAYER: WHAT WAS GROUNDED AND WHAT WAS NOT

Output layer	Made	Written from	Verdict
Title the reader-facing framing of the section	5	Scope	**grounded**
Explanation the principle, rule or relationship to review	5	nothing in the input	**unsourced**
Worked example a problem with its solution worked through	5	nothing in the input	**unsourced**
Figure a drawing the learner reads from	0	nothing in the input	not produced
Practice what the learner does	15	nothing in the input	**unsourced**
Answer what the learner checks against	15	nothing in the input	**unsourced**